PART 2: CALCULATIONS IN CHEMISTRY

CHAPTER 9: Atomic, Molecular and Formula Mass

ATOMIC MASS

In Science it is often necessary to be able to express the mass of an atom. Each atom has a mass of approximately 10^{-23}kg (0.000 000 000 000 000 000 000 01 kg). Masses of atoms in common units (e.g. in kg) are too small to be of practical value hence a relative scale has been devised.

The mass of atoms is expressed using a scale which defines the mass of one atom of the carbon -12 (^{12}C) isotope to be *exactly* 12 atomic mass units.

The average value obtained on this scale for the mass of one atom of an element is termed the RELATIVE ATOMIC MASS of the element.

Relative Atomic Mass (which will be referred to simply as Atomic Mass) was previously referred to as Atomic Weight.

The (RELATIVE) ATOMIC MASS of an element is the ratio of the average mass of an atom of that element to one-twelfth of the mass of one atom of the carbon -12 isotope.

$$\text{Atomic Mass (AM)} = \frac{\text{Average mass of one atom of the element}}{\text{One twelfth mass of one atom of carbon } -12}$$

NOTE: 1. Atomic Mass is a ratio of masses hence it has NO units.
2. As naturally occurring elements generally contain several isotopes, the average mass of an atom compared to one twelfth mass of a carbon -12 atom will produce a ratio which will not be an exact whole number. However, most of the atomic masses given inside the front cover have been rounded to whole numbers.

MOLECULAR MASS, FORMULA MASS

The mass of a molecule of an element or compound can be expressed in similar terms to that used for masses of atoms. The term used is RELATIVE MOLECULAR MASS (or simply MOLECULAR MASS).

The MOLECULAR MASS is the ratio of the average mass of one molecule of a substance to one-twelfth of the mass of a carbon -12 atom.

$$\text{Molecular Mass} = \frac{\text{Average mass of one molecule of substance}}{\text{One twelfth mass of one atom of carbon } -12}$$

The term MOLECULAR MASS was previously referred to as molecular weight (or formula weight). It is used for all substances even though some do not exist in molecular form. The term FORMULA MASS (FM) will be used in this book to avoid this possible confusion.

DETERMINATION OF FORMULA MASS

For any substance (element or compound), the formula mass is a sum of the atomic masses of EACH atom making up the formula of the substance.

The Table of Atomic Masses found inside the cover has been used for all Type Examples and Answers to problems in this book.

TYPE EXAMPLE 1: Find the formula mass of oxygen (gas).

Formula : O_2

$$\begin{aligned} FM(O_2) &= 2 \times AM(O) \\ &= 2 \times 16 \\ &= 32 \end{aligned}$$

i.e. Formula mass of oxygen (gas) is 32.

TYPE EXAMPLE 2: What is the formula mass of sulfur dioxide?

Formula : SO_2

$$\begin{aligned} FM(SO_2) &= AM(S) + 2 \times AM(O) \\ &= 32 + 2(16) \\ &= 64 \end{aligned}$$

i.e. Formula mass of sulfur dioxide is 64.

SET 1: FORMULA MASS I.

Determine the formula mass of each of the following.

1. Water (H_2O)
2. Carbon dioxide (CO_2)
3. Nitrogen gas (N_2)
4. Chlorine gas (Cl_2)
5. Sulfur trioxide (SO_3)
6. Nitrogen dioxide (NO_2)
7. Sulfuric acid (H_2SO_4)
8. Ozone (O_3)
9. Butane (C_4H_{10})
10. Ethanol (C_2H_5OH)

SET 2: FORMULA MASS II.

Write down the formula and determine the formula mass of the following.

1. Hydrogen chloride
2. Calcium oxide
3. Zinc sulfide
4. Aluminium chloride
5. Magnesium oxide
6. Sodium sulfide
7. Potassium bromide
8. Copper(II) oxide
9. Barium chloride
10. Silver sulfide

TYPE EXAMPLE 3: Write down the formula and determine the formula mass of magnesium nitrate.

Formula : $Mg(NO_3)_2$

$$\begin{aligned} FM[Mg(NO_3)_2] &= AM(Mg) + 2\,[AM(N) + 3 \times AM(O)] \\ &= 24 + 2[14 + 3(16)] \\ &= 24 + 2[14 + 48] \\ &= 24 + 2(62) \\ &= 24 + 124 \\ &= 148 \end{aligned}$$

i.e. Formula mass of magnesium nitrate is 148.

SET 3: FORMULA MASS III.

Write down the formula and determine the formula mass of the following.

1. Sodium carbonate
2. Lead (II) nitrate
3. Magnesium hydroxide
4. Copper (II) carbonate
5. Iron (II) hydroxide
6. Ammonium chloride
7. Barium nitrate
8. Potassium hydrogencarbonate
9. Silver nitrate
10. Calcium phosphate

SET 4: FORMULA MASS IV.

Write down the formula and determine the formula mass of the following.

1. Lead (II) hydroxide
2. Potassium sulfate
3. Ammonium sulfide
4. Aluminium carbonate
5. Magnesium nitride
6. Ammonium sulfate
7. Zinc hydrogencarbonate
8. Iron (II) oxide
9. Calcium hydrogensulfate
10. Sodium phosphate

TYPE EXAMPLE 4: Find the formula mass of copper (II) sulfate pentahydrate.

Formula: $CuSO_4.5H_2O$

$$\begin{aligned} FM[CuSO_4.5H_2O] &= AM(Cu) + AM(S) + 4 \times AM(O) + 5[2 \times AM(H) + AM(O)] \\ &= 63.5 + 32 + 4(16) + 5[2(1) + 16] \\ &= 63.5 + 32 + 64 + 5(18) \\ &= 249.5 \end{aligned}$$

i.e. Formula Mass of copper(II) sulfate pentahydrate is 249.5.

SET 5: FORMULA MASS V.

Find the formula mass of each of the following hydrated compounds.

1. Magnesium chloride hexahydrate
2. Zinc chloride monohydrate
3. Iron (II) sulfate heptahydrate
4. Sodium carbonate decahydrate
5. Ferric alum, $NH_4Fe(SO_4)_2.12H_2O$

CHAPTER 11: The Mole

In Science, as previously mentioned, the mass of single atoms is far too small to be of practical value. However, masses must be used in calculations, hence masses based on Atomic (and Formula) Masses are used.

Consider the elements aluminium and carbon —

$$\frac{\text{Atomic mass of aluminium}}{\text{Atomic mass of carbon}} = \frac{27}{12}$$

$$\text{i.e.} \quad \frac{\text{the mass of 1 atom of aluminium}}{\text{the mass of 1 atom of carbon}} = \frac{27}{12}$$

$$\therefore \quad \frac{\text{the mass of 2 atoms of aluminium}}{\text{the mass of 2 atoms of carbon}} = \frac{2 \times 27}{2 \times 12} = \frac{27}{12}$$

$$\text{and} \quad \frac{\text{the mass of 10 atoms of aluminium}}{\text{the mass of 10 atoms of carbon}} = \frac{10 \times 27}{10 \times 12} = \frac{27}{12}$$

$$\therefore \quad \frac{\text{the mass of N atoms of aluminium}}{\text{the mass of N atoms of carbon}} = \frac{N \times 27}{N \times 12} = \frac{27}{12}$$

Whenever the number of aluminium and carbon atoms is the same the ratio of the mass of aluminium to the mass of carbon will be 27:12.

Therefore, if we take 27g of aluminium and 12g of carbon, each sample must contain the same number of atoms.

Similarly, if we take 1g of hydrogen or 14g of nitrogen, we must have the same number of atoms in each sample as there are atoms of carbon in a 12g sample of carbon.

That is, whenever we take a sample of an element which has a mass equal to its Atomic Mass expressed in grams we have the same number of atoms in that sample as there are atoms of carbon in a 12g sample of carbon.

The very large number of atoms contained in such a sample is called Avogadro's Number.

$$\boxed{\text{Avogadro's Number} = 6.02 \times 10^{23} \text{ (approximately)}}$$

Similarly it can be shown that Avogadro's number of molecules of a substance has a mass which is equal to its Formula Mass expressed in grams.

Avogadro's Number (6.02×10^{23}) of particles of a substance is known as a MOLE of that substance.

DEFINITION

There are two acceptable definitions of the MOLE —

1. *A MOLE of a substance is that quantity of that substance which contains AVOGADRO'S NUMBER (6.02×10^{23}) of constituent particles.*
2. *A MOLE of a substance is that quantity of that substance which has a mass equal to its atomic (or formula) mass expressed in grams.*

FROM DEFINITION 1 OF THE MOLE

1 mole of a substance contains 6.02×10^{23} constituent particles.
$\therefore$ n moles of a subtance contains $n \times 6.02 \times 10^{23}$ constituent particles.

i.e. number of particles = number of moles $\times$ (6.02×10^{23})

Rearranging this equation gives,

$$\boxed{\text{number of moles, n} = \frac{\text{number of particles}}{6.02 \times 10^{23}}}$$

TYPE EXAMPLE 11: Calculate the number of moles of ammonia molecules in 60.2×10^{23} molecules of ammonia.

METHOD A (PROPORTION)

1 mole NH_3 contains 6.02×10^{23} molecules
x moles NH_3 contains 60.2×10^{23} molecules

$$\frac{x}{1} = \frac{60.2 \times 10^{23}}{6.02 \times 10^{23}}$$

$$\therefore x = 10 \text{ moles}$$

i.e. 60.2×10^{23} molecules NH_3 contains 10 moles NH_3.

METHOD B (FORMULA)

number of moles of NH_3 = $n(NH_3)$
= ?
number molecules NH_3 = 60.2×10^{23}

$$n(NH_3) = \frac{\text{number molecules } NH_3}{6.02 \times 10^{23}}$$

$$= \frac{60.2 \times 10^{23}}{6.02 \times 10^{23}}$$

$$= 10 \text{ moles}$$

i.e. 60.2×10^{23} molecules NH_3 contains 10 moles NH_3.

SET 13: NUMBER MOLECULES — MOLES

Calculate the number of moles of

1. water molecules in 602×10^{23} molecules water
2. silver atoms in 0.602×10^{23} atoms silver
3. carbon dioxide in 3.01×10^{23} molecules carbon dioxide
4. hydrogen molecules in 120.4×10^{23} molecules hydrogen
5. sulfuric acid in 1.806×10^{23} molecules sulfuric acid

TYPE EXAMPLE 12: Determine the number of molecules of oxygen in 0.4 mole oxygen (O_2).

METHOD A (PROPORTION)

1 mole O_2 contains 6.02×10^{23} molecules
0.4 mole O_2 contains x molecules

$$\frac{0.4}{1} = \frac{x}{6.02 \times 10^{23}}$$
$$\therefore x = 0.4 \times 6.02 \times 10^{23}$$
$$= 2.408 \times 10^{23} \text{ molecules}$$

i.e. 0.4 mole O_2 contains 2.408×10^{23} molecules O_2.

METHOD B (FORMULA)

$n(O_2) = 0.4$ mole

number molecules $O_2 = ?$

$$n(O_2) = \frac{\text{number molecules } O_2}{6.02 \times 10^{23}}$$
$$0.4 = \frac{\text{number molecules } O_2}{6.02 \times 10^{23}}$$
$$\therefore \text{number molecules } O_2 = 0.4 \times 6.02 \times 10^{23}$$
$$= 2.408 \times 10^{23}$$

i.e. 0.4 mole O_2 contains 2.408×10^{23} molecules O_2

SET 13: (continued)

Determine the number of

6. atoms of copper in 0.2 mole copper.
7. molecules of nitrogen dioxide in 5 moles nitrogen dioxide.
8. atoms of carbon in 2 moles carbon.
9. molecules of chlorine in 0.1 mole chlorine (Cl_2).
10. molecules of benzene in 2.5 moles benzene.

FROM DEFINITION 2 OF THE MOLE, to determine the mass of 1 mole of *ANY* substance simply express its formula mass in grams.

TYPE EXAMPLE 13: Determine the mass of 1 mole of —
(a) hydrogen
(b) chlorine atoms
(c) calcium oxide

(a) Hydrogen, H_2:

$$FM[H_2] = 2 \times AM(H)$$
$$= 2 \times 1$$
$$= 2$$

$\therefore$ Mass of 1 mole of hydrogen = 2g.

(b) Chlorine atoms, Cl:

$$AM[Cl] = 35.5$$

$\therefore$ Mass of 1 mole chlorine atoms = 35.5g.

(c) Calcium oxide, CaO:

$$FM[CaO] = AM(Ca) + AM(O)$$
$$= 40 + 16$$
$$= 56$$

$\therefore$ Mass of 1 mole calcium oxide = 56g.

SET 14: MOLE — MASS I

What is the mass of 1 mole of each of the following?

1. Nitrogen (gas)
2. Iron
3. Oxygen atoms
4. Silver bromide
5. Sodium hydroxide
6. Lead (II) oxide
7. Magnesium hydrogensulfate
8. Aluminium hydroxide
9. Zinc carbonate
10. Barium hydrogencarbonate

TYPE EXAMPLE 14: Determine the mass of 0.4 mole magnesium carbonate.

METHOD A (PROPORTION)

Formula: $MgCO_3$

$$FM[MgCO_3] = 24 + 12 + 3(16)$$
$$= 84$$

1 mole magnesium carbonate has mass of 84g
0.4 mole magnesium carbonate has mass of x g

$$\frac{x}{84} = \frac{0.4}{1}$$
$$\therefore x = 84 \times 0.4$$
$$= 33.6g$$

i.e. 0.4 mole $MgCO_3$ has a mass of 33.6g.

METHOD B (FORMULA)

A formula can be derived by simple proportion.

The mass of 1 mole of ANY substance is known as the MOLAR MASS (M) of that substance and is the formula mass expressed in grams.

If the mass of 1 mole = M,
then the mass (m) of n moles = n × mass of 1 mole
= n × M
i.e. m = n × M

This formula is more commonly referred to in the rearranged form–

$$\boxed{n = \frac{m}{M}}$$

where n = number of moles
m = mass of substance (g)
M = molar mass (g)

Note: Strictly speaking the units of M should be $g.mole^{-1}$

Appling this formula to *TYPE EXAMPLE 14.*

METHOD B (FORMULA)

Formula: $MgCO_3$

$FM[MgCO_3] = 24 + 12 + 48$
$= 84$

$n\ (MgCO_3) = 0.4$ mole
$m\ (MgCO_3) = ?$
$M\ (MgCO_3) = 84g$

$$n = \frac{m}{M}$$
$$0.4 = \frac{m}{84}$$
$$\therefore m = 0.4 \times 84$$
$$= 33.6g$$

Again, mass of 0.4 mole $MgCO_3$ is 33.6g.

SET 15: MOLE — MASS II

Determine the mass of —

1. 2 moles aluminium
2. 5 moles oxygen gas
3. 0.2 mole zinc oxide
4. 1.5 moles barium carbonate
5. 0.6 mole copper (II) chloride
6. 0.1 mole iron (II) nitrate
7. 3.2 moles potassium sulfate
8. 2.5 moles sodium hydrogencarbonate
9. 0.5 mole mercury (II) nitrate
10. 0.4 mole iron (II) sulfide

TYPE EXAMPLE 15: Calculate how many moles of water are in 72g water.

Formula: H_2O

$FM[H_2O] = 2(1) + 16 = 18$

Mass of 1 mole water, $M(H_2O) = 18g$

METHOD A (PROPORTION)

1 mole H_2O has a mass of 18g
x mole H_2O has a mass of 72g

$$\frac{x}{1} = \frac{72}{18}$$
$$\therefore x = 4 \text{ moles}$$

i.e. 72g H_2O contains 4 moles H_2O

METHOD B (FORMULA)

$M(H_2O) = 18g$
$m(H_2O) = 72g$
$n(H_2O) = ?$

$$n = \frac{m}{M}$$
$$= \frac{72}{18}$$
$$= 4 \text{ moles}$$

i.e. 72g H_2O contains 4 moles H_2O.

SET 16: MASS — MOLE

Determine the number of moles of

1. magnesium in 48g magnesium
2. hydrogen in 1g hydrogen gas
3. carbon dioxide in 4.4g carbon dioxide
4. calcium sulfate in 27.2g calcium sulfate
5. potassium hydroxide in 22.4g potassium hydroxide
6. zinc nitrate in 947g zinc nitrate
7. ammonium carbonate in 288g ammonium carbonate
8. lead (II) chloride in 55.6g lead (II) chloride
9. sodium sulfate in 85.2g sodium sulfate
10. silver chloride in 114.8g silver chloride

TYPE EXAMPLE 16; If 1.5 moles of benzene has a mass of 117g, determine the formula mass of benzene.

METHOD A (PROPORTION)

1.5 moles has mass of 117g.
1 mole benzene has mass of xg

$$\frac{x}{117} = \frac{1}{1.5}$$
$$\therefore x = \frac{117}{1.5}$$
$$= 78g$$

From the definition of a mole, the mass of 1 mole of any substance is its formula mass expressed in grams.

$\therefore$ Formula mass of benzene is 78

METHOD B (FORMULA)

n (benzene) = 1.5 moles
m (benzene) = 117g
M (benzene) = ?

$$n = \frac{m}{M}$$
$$1.5 = \frac{117}{M}$$
$$\therefore M = \frac{117}{1.5}$$
$$= 78g$$

i.e. Formula mass of benzene is 78.

SET 17: FORMULA MASS

Determine the formula mass of the substance in each of the questions below, if

1. 3 moles acetylene has a mass of 78g
2. 2 moles phenol has a mass of 188g
3. 0.1 mole calcium hypochlorite has a mass of 14.3g
4. 0.2 mole methanol has a mass of 6.4g
5. 1.5 moles ethylene has a mass of 42g
6. the mass of 5 moles propane is 220g
7. the mass of 0.4 mole ethanol is 18.4g
8. the mass of 2.5 moles of a gas is 115g
9. the mass of 0.1 mole ethane is 3g
10. the mass of 0.3 mole of a silvery metal is 15.6g.

TYPE EXAMPLE 17: Determine the number of moles of atoms of hydrogen, sulfur and oxygen in 1 mole sulfuric acid.

Formula: H_2SO_4

Examination of the formula shows 1 molecule of H_2SO_4 contains–

(a) 2 atoms H
(b) 1 atom S
(c) 4 atoms O

$\therefore$ Avogadro's Number of molecules of H_2SO_4 must contain —

(a) 2 × Avogadro's Number of H atoms.
(b) 1 × Avogadro's Number of S atoms.
(c) 4 × Avogadro's Number of O atoms.

i.e. 1 mole H_2SO_4 contains

(a) 2 moles H atoms
(b) 1 mole S atoms
(c) 4 moles O atoms

TYPE EXAMPLE 18: Find the number of moles of potassium ions in 0.2 mole potassium sulfate.

Formula: K_2SO_4

MASS-MASS DETERMINATIONS:

TYPE EXAMPLE 32: What mass of hydrogen is produced when 6g magnesium is added to excess sulfuric acid?

METHOD A (PROPORTION)

$$\underset{1\ \text{mole}}{Mg} + H_2SO_4 \longrightarrow MgSO_4 + \underset{1\ \text{mole}}{H_2}$$

FM (Mg) = 24
FM (H_2) = 2

1 mole magnesium produces 1 mole hydrogen (from equation)
∴ 24g magnesium produces 2g hydrogen
6g magnesium produces xg hydrogen

$$\frac{x}{2} = \frac{6}{24}$$
$$\therefore x = \frac{2 \times 6}{24}$$
$$= 0.5g$$

i.e. 6g Mg produces 0.5g H_2.

METHOD B (FORMULA)

Step 1:

$$\underset{\text{(known)}}{Mg} + H_2SO_4 \longrightarrow MgSO_4 + \underset{\text{(unknown)}}{H_2}$$

FM(Mg) = 24
FM(H_2) = 2

Step 2:

m(Mg) = 6g
n(Mg) = ?
M(Mg) = 24g

$$n = \frac{m}{M}$$
$$= \frac{6}{24}$$
$$= 0.25\ \text{mole}$$

Step 3:

$$\frac{n(H_2)}{n(Mg)} = \frac{1}{1}$$

Step 4:

n(Mg) = 0.25 mole
n(H_2) = ?

$$\frac{n(H_2)}{0.25} = \frac{1}{1}$$
$$\therefore n(H_2) = 0.25\ \text{mole}$$

Step 5:

n(H_2) = 0.25 mole
m(H_2) = ?
M(H_2) = 2g

$$n = \frac{m}{M}$$
$$0.25 = \frac{m}{2}$$
$$\therefore m = 0.25 \times 2$$
$$= 0.5g$$

i.e. 6g Mg produces 0.5g H_2.

SET 32: MASS-MASS DETERMINATIONS I

1. 25g calcium carbonate is strongly heated.
   $$CaCO_3 \longrightarrow CaO + CO_2$$
   Determine the mass of calcium oxide produced.

2. In an experiment 3.6g water is electrolysed.
   $$2H_2O \longrightarrow 2H_2 + O_2$$
   Find the mass of
   (a) hydrogen, and
   (b) oxygen produced.

3. 20g sodium hydroxide is neutralized by dilute hydrochloric acid.
   $$NaOH + HCl \longrightarrow NaCl + H_2O$$
   How many grams of sodium chloride can be recovered?

4. Determine the mass of calcium sulfate produced when 5.6g calcium oxide is neutralized by sulfuric acid.
   $$CaO + H_2SO_4 \longrightarrow CaSO_4 + H_2O$$

5. 6.9g sodium metal is reacted with water. Find the mass of sodium hydroxide produced.
   $$2Na + 2H_2O \longrightarrow 2NaOH + H_2$$

6. 420g sodium hydrogencarbonate is strongly heated.
   $$2NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$$
   Determine the mass of —
   (a) sodium carbonate left, and
   (b) carbon dioxide produced.

7. Determine the mass of hydrochloric acid required to completely react with 16.8g magnesium carbonate.
   $$MgCO_3 + 2HCl \longrightarrow MgCl_2 + H_2O + CO_2$$

8. 6.35g copper is dissolved when placed in a solution of silver nitrate.

$$Cu(s) + 2AgNO_3 \longrightarrow Cu(NO_3)_2 + 2Ag(s)$$

How many grams of
(a) silver is deposited, and
(b) copper(II) nitrate can be recovered?

9. Sufficient potassium carbonate solution is added to 34.1g zinc chloride to completely precipitate the zinc as zinc carbonate.

$$ZnCl_2 + K_2CO_3 \longrightarrow ZnCO_3(s) + 2KCl$$

What mass of zinc carbonate is deposited?

10. Determine the mass of
(a) sodium nitrite left, and
(b) oxygen gas evolved
when 17g sodium nitrate is strongly heated

$$2NaNO_3 \longrightarrow 2NaNO_2 + O_2(g)$$

SET 33: MASS-MASS DETERMINATIONS II

1. Determine the mass of potassium chlorate which must be decomposed by heating to produce 9.6g oxygen.

$$2KClO_3 \longrightarrow 2KCl + 3O_2$$

2. 10.8g aluminium powder is burnt in air.

$$4Al + 3O_2 \longrightarrow 2Al_2O_3$$

Find the mass of aluminium oxide produced.

3. 66g ammonium sulfate is produced when ammonia gas is bubbled through a sulfuric acid solution.

$$2NH_3(g) + H_2SO_4 \longrightarrow (NH_4)_2SO_4.$$

What mass of ammonia is required?

4. Calculate the mass of hydrogen peroxide which upon catalytic decomposition will produce 4g oxygen.

$$2H_2O_2 \longrightarrow 2H_2O + O_2$$

5. Determine the mass of sodium carbonate which must be added to a calcium nitrate solution to precipitate all of the calcium as 8g of calcium carbonate.

$$Na_2CO_3 + Ca(NO_3)_2 \longrightarrow CaCO_3(s) + 2NaNO_3$$

6. Find the mass of chlorine produced when 4.35g manganese dioxide is reacted with concentrated hydrochloric acid.

$$MnO_2 + 4HCl \longrightarrow MnCl_2 + 2H_2O + Cl_2$$

7. What mass of barium chloride is produced in the reaction between hydrochloric acid and barium carbonate which produces 8.8g carbon dioxide?

$$BaCO_3 + 2HCl \longrightarrow BaCl_2 + H_2O + CO_2(g)$$

8. Determine the mass of carbon dioxide formed when 16kg methane (CH_4) burns in air.

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$$

9. 71.5g sodium carbonate decahydrate is heated until it is completely dehydrated. Calculate the mass of water (as steam) driven off.

$$Na_2CO_3.10H_2O \longrightarrow Na_2CO_3 + 10H_2O(g)$$

10. A sample of mercury(II) oxide on heating produced 1.6g oxygen gas.

$$2HgO \longrightarrow 2Hg + O_2(g)$$

Determine
(a) the mass of mercury(II) oxide used, and
(b) the mass of mercury remaining.

MASS-VOLUME DETERMINATIONS:

In Chapter II the term MOLAR VOLUME was defined as the volume occupied by 1 mole of a gas.

It was shown that the Molar Volume for any gas is 22.4L when measured under S.T.P. conditions.

The formula for mole — volume determinations was derived:

$$n = \frac{V}{\text{Molar Volume}} \quad \text{(any conditions)}$$

OR

$$n = \frac{V}{22.4}$$

Where
n = moles of gas
V = volume of gas in litres
22.4L = Molar Volume (at S.T.P.)

TYPE EXAMPLE 33: What volume of sulfur dioxide is produced (at S.T.P.) by burning 3.2g sulfur?

METHOD A (PROPORTION)

$$S_8 + 8O_2 \longrightarrow 8SO_2(g)$$

1 mole 8 moles

$FM(S_8) = 8(32) = 256$

ANSWERS — PART 2

SET 1

1. 18 **2.** 44 **3.** 28 **4.** 71 **5.** 80
6. 46 **7.** 98 **8.** 48 **9.** 58 **10.** 46

SET 2

1. HCl, 36.5 **2.** CaO, 56 **3.** ZnS, 97.4
4. $AlCl_3$, 133.5 **5.** MgO, 40 **6.** Na_2S, 78
7. KBr, 119 **8.** CuO, 79.5 **9.** $BaCl_2$, 208
10. Ag_2S, 248

SET 3

1. Na_2CO_3, 106 **2.** $Pb(NO_3)_2$, 331 **3.** $Mg(OH)_2$, 58
4. $CuCO_3$, 123.5 **5.** $Fe(OH)_2$, 90 **6.** NH_4Cl, 53.5
7. $Ba(NO_3)_2$, 261 **8.** $KHCO_3$, 100 **9.** $AgNO_3$, 170
10. $Ca_3(PO_4)_2$, 310

SET 4

1. $Pb(OH)_2$, 241 **2.** K_2SO_4, 174 **3.** $(NH_4)_2S$, 68
4. $Al_2(CO_3)_3$, 234 **5.** Mg_3N_2, 100 **6.** $(NH_4)_2SO_4$, 132
7. $Zn(HCO_3)_2$, 187.4 **8.** FeO, 72 **9.** $Ca(HSO_4)_2$, 234
10. Na_3PO_4, 164

SET 5

1. 203 **2.** 154.4 **3.** 278 **4.** 286 **5.** 482

SET 6

1. C 27.3%; O 72.7% **2.** N 82.4%; H 17.6%
3. Ca 40%; C 12%; O 48% **4.** Mg 25.3%; Cl 74.7%
5. Ag 63.5%; N 8.2%; O 28.2%

SET 7

1. 11.1% **2.** 40% **3.** 75% **4.** 57.1% **5.** 32.4%
6. 18.9% **7.** 74.5% **8.** 18.7% **9.** 92.6% **10.** 23.5%

SET 8

1. 72.9% **2.** 22.5% **3.** 47.7% **4.** 84.2% **5.** 83.6%

SET 9

1. 14.8% **2.** 51.2% **3.** 21.1% **4.** 39.9% **5.** 22.4%

SET 10

1. 4g **2.** 6.4g **3.** 2.8g **4.** 254g **5.** 130.8g
6. 68.5g **7.** 1.2g **8.** 6.9g **9.** 1g **10.** 8.4g

SET 11

1. 14g **2.** 117g **3.** 11.6g **4.** 51g **5.** 26.4g
6. 46.1g **7.** 116.5g **8.** 44.7g **9.** 35.6g **10.** 8.7g

SET 12

1(a) $CaCl_2$, 111 **(b)** NH_4NO_3, 80
2(a) $Ba(HSO_4)_2$, 331 **(b)** $Zn(OH)_2$, 99.4
3. Mg, 20%; S, 26.7%; O, 53.3% **4.** 8.7%
5. 62.9% **6.** 6.4g **7.** 32.7g
8. 39g **9.** 89g **10.** 54.25g

SET 13

1. 100 **2.** 0.1 **3.** 0.5 **4.** 20 **5.** 0.3
6. 1.204×10^{23} **7.** 30.1×10^{23} **8.** 12.04×10^{23}
9. 0.602×10^{23} **10.** 15.05×10^{23}

SET 14

1. 28g **2.** 56g **3.** 16g **4.** 188g **5.** 40g
6. 223g **7.** 218g **8.** 78g **9.** 125.4g **10.** 259g

SET 15

1. 54g **2.** 160g **3.** 16.28g **4.** 295.5g **5.** 80.7g
6. 18g **7.** 556.8g **8.** 210g **9.** 162.5g **10.** 35.2g

SET 16

1. 2 **2.** 0.5 **3.** 0.1 **4.** 0.2 **5.** 0.4
6. 5 **7.** 3 **8.** 0.2 **9.** 0.6 **10.** 0.8

SET 17

1. 26 **2.** 94 **3.** 143 **4.** 32 **5.** 28
6. 44 **7.** 46 **8.** 46 **9.** 30 **10.** 52

SET 18

1. 0.2 **2.** H, 2; N, 2; O, 6 **3.** Cu, 2.5; S, 2.5; O, 10
4. Ca^{2+}, 0.4; OH^-, 0.8 **5.** 0.602×10^{23} of each
6. H, 36.12×10^{23}; S, 18.06×10^{23}
7.(a) N, 0.2; H, 0.6 **(b)** NH_3, 1.204×10^{23}; N, 1.204×10^{23}; H, 3.612×10^{23}
8. N, 1; H, 4; C, 0.5; O, 1.5

SET 19

1. 0.1 2. 0.4 3. 2 4. 1.5 5. 0.4
6. 3 7. 0.15 8. 0.75 9. 0.4 10. 1.4

SET 20

1. 32g 2. 4g 3. 14g 4. 6.4g 5. 22.4g
6. 48g 7. 11.7g 8. 2 010g 9. 12.8g 10. 48g

SET 21

1.(a) 40g (b) 73g 2.(a) 0.602×10^{23} (b) 36.12×10^{23}
3.(a) 0.2 (b) 0.4 4. 0.5 5.(a) 28g (b) 14g
6. 249.5g 7.(a) 46.4g (b) 390g 8. 0.3
9.(a) 0.1 (b) 1 10. 60g 11.(a) 0.2 (b) 2.8g
12. 70 13.(a) 255g (b) 53g 14. 0.29
15. 5.36 16. 0.4 17.(a) 106.5g (b) 7.1g
18. 16.8g 19. 0.4 20. 64

SET 22

1. 0.2 mol L^{-1} 2. 165.5 g L^{-1} 3. 50 g L^{-1}, 1.25 mol L^{-1} 4. 0.25 mol L^{-1}
21.25 g L^{-1} 5. 0.279 g L^{-1}, 2.56 g L^{-1} 6. 6.15 mol L^{-1}, 0.0216 mol L^{-1}
7. 4.56 mol L^{-1}, 7.61 mol L^{-1} 8. 0.0341 mol L^{-1}, 0.0204 mol L^{-1} 9. 0.0384 g L^{-1},
0.0736 g L^{-1} 10. (a) about 53^0 (b) (i) about 525 g L^{-1} (ii) about 310gL^{-1}
(c) 215g (d) about 0.88 mol L^{-1} (e) about 32^0C

SET 23

1. 156.8L 2. 56L 3. 0.896L 4. 224L 5. 0.0224L (22.4mL)
6. 0.1 7. 0.05 8. 8 9. 70 10. 30

SET 24

1. 44.8L 2. 5.6L 3. 8.96L 4. 89.6L 5. 1.12L
6. 4.48L 7. 33.6L 8. 2.8L 9. 24.64L 10. 56L

SET 25

1. 11g 2. 3.55g 3. 0.6g 4. 160g 5. 10g
6. 40g 7. 127.75g 8. 2.55g 9. 34.5g 10. 3.2g

SET 26

1. 53.76L 2. 2.5 3. 4.48L 4. 1.4L 5. 38.4g
6.(a) 7.5 (b) 330g 7. 0.4 8. 15.68L 9. 0.75 10. 45.6g

SET 27 qualitative, no answers

SET 28

1. 4 2.(a) 0.4 (b) 0.2 3. 0.2 4.(a) 0.3 (b) 0.3 5.(a) 1.5 (b) 0.75
6. 1 7. 0.22 8. 0.02 9. 0.2 10. 0.4

SET 29

1. 102.4g 2.(a) 5.3g (b) 2.2g 3. 17.75g
4.(a) 14.9g (b) 9.6g 5. 0.72g

SET 30

1. 4.8g 2.(a) 68g (b) 196g 3. 8.7g
4.(a) 6.175g (b) 2.2g 5.(a) 31.2g (b) 0.8

SET 31

1. 0.2 2. 0.5 3.(a) 0.1 (b) 0.05
4. 0.4 5. 0.25 6.(a) 0.25 (b) 0.25
7. 0.5 8. 8 9. 0.4
10. 0.08

SET 32

1. 14g 2.(a) 0.4g (b) 3.2g 3. 29.25g
4. 13.6g 5. 12g 6.(a) 265g (b) 110g
7. 14.6g 8.(a) 21.6g (b) 18.75g 9. 31.35g
10.(a) 13.8g (b) 3.2g

SET 33

1. 24.5g 2. 20.4g 3. 17g 4. 8.5g 5. 8.48g
6. 3.55g 7. 41.6g 8. 44kg 9. 45g
10.(a) 21.7g (b) 20.1g

SET 34

1. 4.48L 2. 1.12L 3. 0.224L
4.(a) 5.6L (b) 2.8L 5. 2.8L 6.(a) 1.12L (b) 6.9g
7. 3.36L 8.(a) 2.24L (b) 6.72L 9.(a) 336L (b) 1375g
10. 2.24L

SET 35

1. 0.2 2. 18g 3. 14g
4.(a) 10.7g (b) 7.4g 5. 7.1kg 6.(a) 4 (b) 400g
7. 9.6g 8. 15.6g 9. 64.35g
10.(a) 3.175g (b) 9.375g